

B.E. Even Semester Midterm Examination, 2024-25

Subject: Data Structures / Design and Analysis of Algorithms

Q1. Find the MST from the following graph using Kruskal algorithm

(Graph Edges: $AB=1$, $AC=2$, $BC=3$, $CD=4$, $CE=5$, $CF=6$, $DE=7$, $EF=8$) **Answer:** Kruskal's Algorithm finds the Minimum Spanning Tree (MST) by sorting edges and adding them avoiding cycles. (5 edges needed for 6 vertices). Sorted: $AB(1)$, $AC(2)$, $BC(3)$, $CD(4)$, $CE(5)$, $CF(6)$, $DE(7)$, $EF(8)$. 1. Add AB (1) 2. Add AC (2) 3. Reject BC (creates cycle $A-B-C$) 4. Add CD (4) 5. Add CE (5) 6. Add CF (6) **MST Edges:** AB, AC, CD, CE, CF . **Total Weight:** 18.

Q2. Breadth first traversal of a graph uses representation of graph in which data structure? Find the BFS sequence for the following graph.

Answer: * **Data Structure:** Breadth-First Search utilizes a **Queue** (First-In-First-Out) data structure. * **BFS Sequence:** Assuming start at M: Queue: [M]. Output: M. Enqueue N, Q, R. Queue: [N, Q, R]. Dequeue N. Output: N. Enqueue O. Queue: [Q, R, O]. Dequeue Q. Output: Q. Enqueue P. Queue: [R, O, P]. Dequeue R. Output: R. Queue: [O, P]. Dequeue O. Output: O. Queue: [P]. Dequeue P. Output: P. **Sequence:** M, N, Q, R, O, P.

Q3. Why are all Master's theorems compared with the function $\$n^{\log_b(a)}\$$?

Answer: In the recurrence $\$T(n) = aT(n/b) + f(n)\$,$ expanding it creates a recursion tree. $\$n^{\log_b(a)}\$$ represents the total amount of work done at the leaf level (base cases) of the tree. The function $\$f(n)\$$ represents the work done at the root node (combining). Master's theorem compares $\$f(n)\$$ with $\$n^{\log_b(a)}\$$ to determine which part of the tree dominates the computational cost (leaves, root, or evenly balanced).